

Nursing the cat with chronic kidney disease

Chronic kidney disease (CKD) is a common cause of illness in cats, particularly as they age. Research over the past 10 years has advanced our understanding of the condition, with resultant developments for slowing progression and managing clinical signs. This article provides some background on the causes of CKD and information about its diagnosis and management, and demonstrates the vital role veterinary nurses and technicians can play in all aspects of caring for cats with CKD.



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Functions of the kidney

The kidney is involved in multiple physiological processes in the body and is key for homeostasis. Its diverse roles explain the impact of renal dysfunction on the individual cat. Key functions include:

- Maintaining fluid balance and electrolyte concentrations;
- Acid-base balance;
- Red blood cell formation via erythropoietin production;
- Production of renin as part of the renin-angiotensin-aldosterone system (RAAS) (blood pressure regulation);
- Calcium and phosphate balance, and hence bone metabolism.

Due to the potential effects of CKD on many physiological processes, management involves addressing multiple areas to reduce clinical signs of illness and slow progression of the disease.

Pathophysiology of CKD

CKD has a prevalence of 28–81% in all cats aged >12 years, but can occur at any age.^{1–5} In the majority of cats, the underlying cause of their CKD cannot be determined, and, if renal biopsies are performed, histopathology reveals chronic tubulointerstitial nephritis of unknown

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aetiology (Figure 1).⁶ These changes are irreversible, but in many cats the progress of the disease is slow. However, early diagnosis may allow prompt interventions to both slow progress further and improve clinical signs. Causes of CKD that can be identified include:

- Congenital and hereditary causes
 - Polycystic kidney disease (Figure 2)
 - Inherited amyloidosis
 - Juvenile nephropathy
- Glomerular disease
 - Immune complex glomerulonephritis
- Obstructive and postrenal causes
 - Ureterolithiasis
 - Ureteral strictures
 - Neoplasia causing obstruction
 - Ureteral stenosis
 - Hydronephrosis
- Infectious causes
 - Bacterial pyelonephritis
 - Viruses: feline leukaemia virus, feline infectious peritonitis
 - Parasites: *Diocotophyme renale*
- Toxic causes (chronic or delayed effects)
 - Plants: lilies
 - Drugs: non-steroidal anti-inflammatory drugs, chemotherapeutics, aminoglycosides
 - Chemicals: ethylene glycol, heavy metals (eg, lead)
- Vascular and haemodynamic causes
 - Systemic hypertension (via proteinuria)
 - Renal ischaemia
 - Thromboembolic disease
- Endocrine and metabolic disorders
 - Hyperthyroidism
 - Primary hyperaldosteronism
 - Diabetes mellitus
 - Hypercalcaemia
- Neoplastic causes
 - Renal lymphoma
 - Renal carcinoma
 - Infiltrative neoplasia

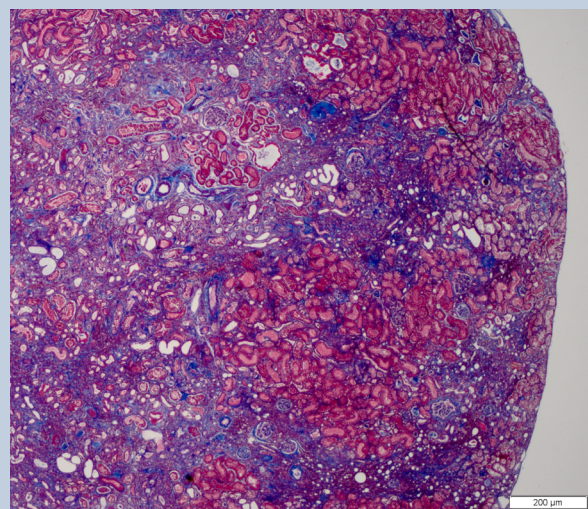


Figure 1: Histology image (trichrome stain) of tubulointerstitial nephritis, the typical change found in cats with chronic kidney disease, illustrating focal areas of interstitial inflammation, fibrosis and tubular loss (blue, purple) interspersed among healthy tubules (pink). *Courtesy of Shannon McLeland*

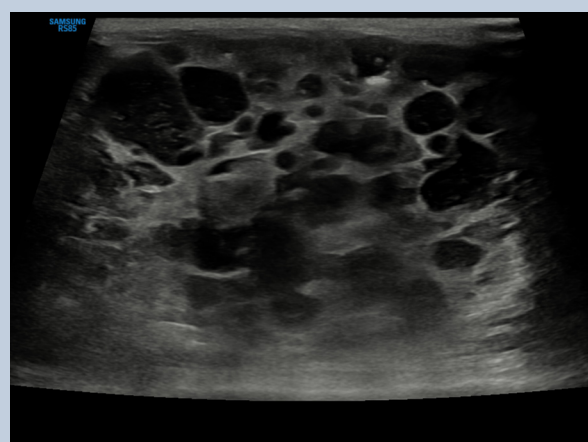


Figure 2: Ultrasound image of a polycystic kidney in a Persian cat. *Courtesy of Lumbry Park Veterinary Specialists*

- Iatrogenic causes
 - Repeated anaesthesia with hypotension
 - Chronic nephrotoxin exposure
 - Radiation therapy involving the kidneys
- Sequela of acute kidney injury

Progression of CKD

Several factors are associated with CKD progression in cats: the severity

of proteinuria is strongly associated with survival time,⁷ and increased plasma phosphate concentration is associated with progression within 12 months of diagnosis,⁸ as well as with a short survival time.⁹ Other biomarkers such as fibroblast growth factor-23 may have prognostic significance.¹⁰ Monitoring in the first 3 months from diagnosis can aid prediction of progression, suggesting that serial monitoring of creatinine, phosphate and body weight can identify cats with progressive CKD vs stable disease, with progressive CKD cats having a significantly shorter survival time (>60% dying within 365 days of diagnosis).¹¹ Other factors associated with progression of CKD include anaemia,⁸ uraemic toxins¹² and hypokalaemia (the last in humans with CKD).¹³

Clinical signs of CKD

Clinical signs associated with CKD are typically insidious in onset and often non-specific. The most commonly reported signs are polyuria, polydipsia and weight loss, followed by loss of appetite and vomiting.¹⁴⁻¹⁶ Other common signs include lethargy and weakness, although many cats show no clinical signs and may only be diagnosed via health screening.¹⁶⁻¹⁸ Weight loss (Figure 3), typically gradual in onset, is commonly reported in cats with CKD,^{14,19} and may occur due to nausea and malaise associated with complications of CKD such as hypokalaemia, anaemia, acidosis, constipation and dehydration, and associated appetite dysregulation due to anorexogenic substances.²⁰ This insidious weight loss and loss of muscle mass may go unnoticed by caregivers or be incorrectly attributed to age. Veterinary nurses and technicians can promote monitoring of body weight as well as body condition score (BCS) and muscle condition score (MCS) in

at-risk populations (eg, older cats) to improve early identification of body composition and weight changes that could indicate underlying conditions such as CKD. Unintended weight loss or a deterioration in BCS or MCS should not be ignored and should prompt consideration of further diagnostic testing. Health clinics for senior cats provide an opportunity to detect cases of CKD and educate caregivers on clinical signs to be vigilant for in their cats.

Less frequent signs of CKD include weakness and neck ventroflexion due to hypokalaemia²¹ and lower urinary tract signs due to urinary tract infection and constipation.^{22,23} Constipation may indicate chronic dehydration and/or hypokalaemia and asking caregivers about the frequency of their cat's defecation and the consistency of their faeces is therefore important. Furthermore, uraemic halitosis and oral ulceration may develop in cats with advanced CKD.¹⁵



Figure 3: Weight loss is common in cats with chronic kidney disease (CKD) and may be the only clinical sign. This patient also has a poor haircoat and reduced muscle condition, both of which are also typical findings in cats with CKD. *Courtesy of Sarah Caney*

Physical examination for CKD

Physical examination may be unremarkable in cats with CKD, but may identify pallor in anaemic cats, evidence of dehydration (dry mucous membranes, prolonged skin tent) and kidneys may feel small and firm. Firm faeces may be palpable in the colon and a reduced BCS and MCS may be noted. It is important to measure blood pressure in cats with CKD (Figure 4) as hypertension is common in this group of cats; affected cats may also have ocular changes such as hyphaema and retinal detachment (Figure 5). Cats with CKD may have comorbid conditions that may be detected on clinical examination; for example, goitre in hyperthyroidism.

Diagnostics for CKD

In most cases, a diagnosis of CKD is based on a creatinine concentration above the laboratory reference interval (RI) and/or evidence of structural change on imaging or renal palpation, with consideration of other supportive findings such as an inappropriately low urine specific gravity (USG <1.035) and/or proteinuria. Importantly, cats with CKD can have serum creatinine concentrations within the laboratory RI, as the upper limit in older cats should be lower than the general population and serum creatinine is highly individual. Monitoring trends in an individual cat to identify increases in creatinine over time can be extremely useful in making an early diagnosis of CKD.

Serum creatinine and SDMA

Creatinine is the most widely used surrogate marker of glomerular filtration rate (GFR). It is the preferred marker over urea as it is less affected by non-renal factors such as dehydration or food consumption.²⁴ However, creatinine concentration has its limitations: it is influenced by muscle mass and



Figure 4: Measurement of blood pressure is important in cats with chronic kidney disease to identify those with hypertension. *Courtesy of Sam Taylor*



Figure 5: A cat with chronic kidney disease and hypertension with acute-onset blindness, with bilaterally dilated pupils and notable hyphaema in the left eye. *Courtesy of Sam Taylor*

hydration status and should be interpreted with physical examination findings. Symmetric dimethylarginine (SDMA) is a more recently available biomarker of GFR and is less affected by muscle mass than creatinine but it is still influenced by hydration and comorbidities.²⁵

Other clinical pathology findings

Various other abnormalities may be detected in cats with CKD, such as low or high calcium, hyperphosphataemia, hypokalaemia and anaemia. Normocytic normochromic non-regenerative anaemia is common in cats with CKD,^{6,8} especially in the later stages

of the disease. Thus, assessment of a complete blood count, or, as a minimum, a packed cell volume (PCV) and blood smear review, is indicated. Cats with pyelonephritis may have a leukocytosis.

Urinalysis

The most important urinary parameters to evaluate as part of the diagnostic work-up include USG and urine protein. One of the earliest manifestations of decreased kidney function is a reduction in urine-concentrating ability (USG <1.035).^{26,27} In many cases, urine-concentrating ability decreases or is lost before serum creatinine exceeds the upper limit of the RI.¹⁵ However, despite urine concentration being one of the earliest markers of decreased kidney function, cats with early-stage CKD and mildly elevated creatinine could have appropriately concentrated urine (ie, USG ≥ 1.035). Conversely, USG <1.035 could be appropriate in normal cats on diets with a high water content, for example. Hence, a single measurement of this parameter is less useful than repeated measurements in an individual cat.²⁴

Proteinuria is another valuable and clinically relevant parameter in the diagnosis and evaluation of feline CKD. Urine sediment examination can also provide clinically relevant

information that may either identify the cause or highlight complications of the disease (eg, crystalluria, bacteriuria) and is thus an additional essential aspect of urinalysis in cats with suspected CKD.

Diagnostic imaging

Diagnostic imaging is an important part of the assessment of cats presenting with azotaemia or suspected renal disease. The aim is to rule in intrinsic renal disease as well as rule out postrenal azotaemia (ie, ureteral obstruction and/or hydronephrosis; Figure 6) or other causes of azotaemia such as renal neoplasia,²⁸ polycystic kidney disease or infectious causes (eg, feline infectious peritonitis, pyelonephritis).²⁹ The kidneys of cats with CKD are small and irregular, with loss of corticomedullary definition.



Figure 6: Ultrasound image of a cat with ureteral obstruction and hydronephrosis. *Courtesy of Sam Taylor*

'Staging' of CKD: why is it important?

The International Renal Interest Society (IRIS), a panel of expert veterinarians, has produced guidance to assist veterinary practitioners in diagnosing and treating kidney diseases in dogs and cats. Central to this is an established method of 'staging' CKD based initially on a stable creatinine or SDMA measurement in a hydrated patient. By assigning a 'stage' – from 1 (early/mild disease) to 4 (severe disease) – appropriate treatment and monitoring can be facilitated in cats with CKD and caregivers provided with a prognosis. A further process of 'substaging' is then recommended, based on both the cat's blood pressure and amount of protein in the urine. A cat's CKD stage will progressively increase over time, so regular reassessments should be carried out to re-stage the patient. Cats at higher stages of disease are likely to have more clinical signs and require more intervention to maintain their quality of life. See www.iris-kidney.com for more information.

Management of CKD

In broad terms, the management of CKD aims to address clinical signs to optimise the quality of life of the affected cat and slow progression of the disease. Working as a team with the caregiver is essential to ensure interventions are appropriate for the individual situation (Figure 7). Veterinary nurses and technicians should be involved in all aspects of management of a cat with CKD including in-patient care, nutritional assessment and support, and communication with caregivers on medicating and monitoring.

Dietary interventions

There is strong evidence to support dietary modification in cats with CKD. Feeding a veterinary therapeutic renal diet can prolong survival and reduce the risk of uraemic crises developing in cats with CKD.^{30,31} Renal diets may

also have other benefits for cats with CKD – for example, supplementation with omega 3 fatty acids, such as eicosapentaenoic acid (EPA), as well as potassium and B vitamins; they may also contain alkalinising ingredients. When it comes to choosing a specific reduced-phosphate diet for an individual cat, a nutritional assessment should be performed, including measurement of body weight, BCS and MCS, as well as identification of cat preferences (flavour, texture, frequency of feeding) and relevant caregiver factors (finances, lifestyle) to allow selection of a diet more likely to be offered by the caregiver and accepted by the cat. Other considerations include:

- Feeding a canned or moist diet will provide more water to the cat, which can aid in maintaining adequate hydration.

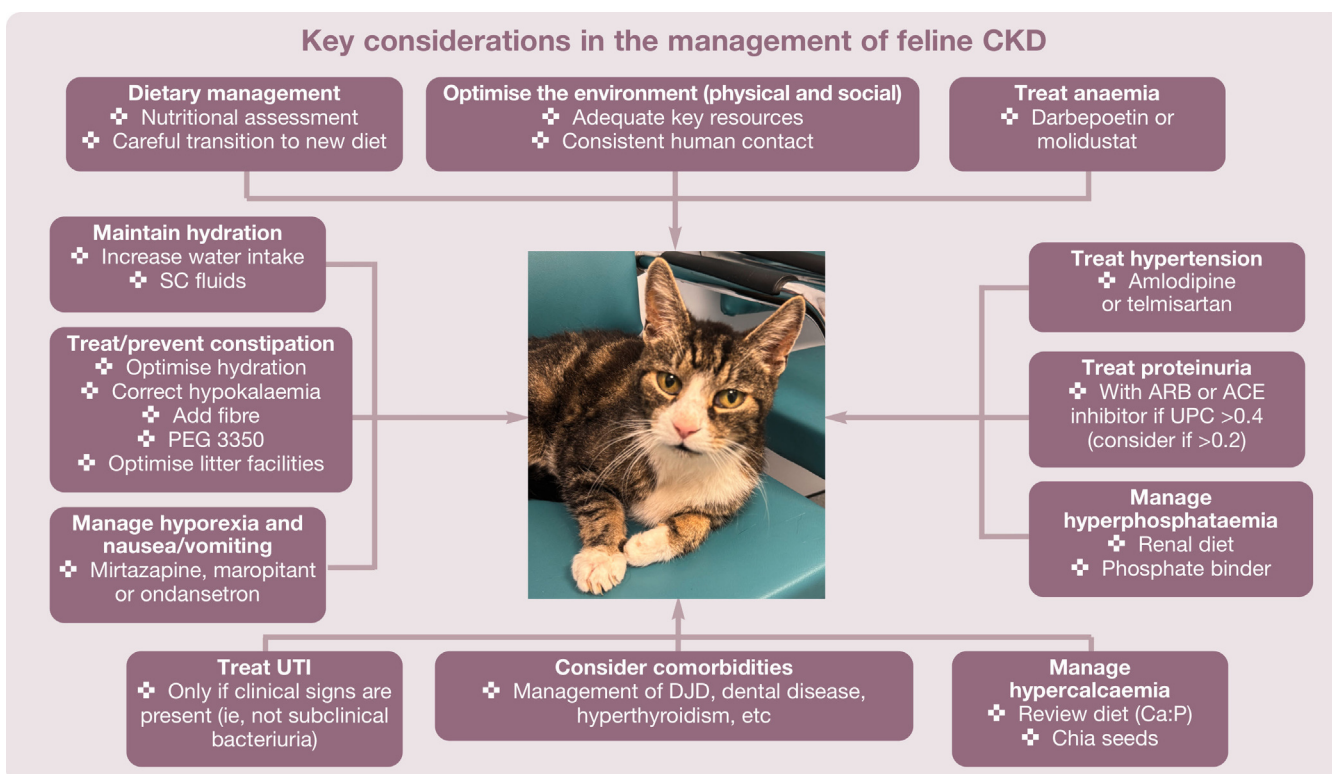


Figure 7: Factors to consider in the management of cats with chronic kidney disease (CKD). ACE = angiotensin-converting enzyme; ARB = angiotensin receptor blocker; Ca:P = calcium:phosphorus ratio; DJD = degenerative joint disease; PEG = polyethylene glycol; SC = subcutaneous; UPC = urine protein:creatinine ratio; UTI = urinary tract infection; 2026 iCatCare consensus guidelines on the diagnosis and management of chronic kidney disease in cats. *J Feline Med Surg*. Available at: <https://journals.sagepub.com/doi/full/10.1177/1098612X261466553>. Copyright The Authors 2026

- Canned diets will be lower in caloric density, requiring cats to eat a greater volume of food, and it may be challenging for some cats to maintain weight, whereas kibble or dry diets will typically provide about four times more calories per gram than most canned or fresh diets.
- Cats with CKD can have a waxing-waning appetite; hence, having multiple appropriate diet options that meet nutritional needs is often desirable.
- Ideally, a therapeutic renal diet would be fed exclusively; however, this is not always possible, so caregivers should be encouraged to continue to feed the renal diet even if it only makes up a proportion of the calories eaten. A renal diet comprising 50% or more of the cat's intake can have a beneficial effect on survival.³⁰
- Some cats will maintain weight simply by eating their resting energy requirement (RER; calculated as $70 \times \text{body weight [kg]}^{0.75}$). However, others will require their RER to be multiplied by a maintenance energy requirement factor of approximately 1.2–1.4. If a cat requires more than 2.0 RER, additional diagnostics should be performed to assess for comorbid conditions (eg, hyperthyroidism, chronic enteropathy, exocrine pancreatic insufficiency).

When introducing a new diet to a cat with CKD, it is best to offer small amounts of the new food gradually over at least 1–2 weeks alongside the existing food. Cats often do not like having the new and existing diets mixed, so offering the foods side by side may be better tolerated. It is important never to introduce a novel diet during a period of illness or, for example, while a cat is undergoing diagnostic investigations within the veterinary clinic, as this could lead to

Key point

Cats with chronic kidney disease can have a waxing-waning appetite; hence, having multiple appropriate diet options that meet nutritional needs is often desirable.



the development of food aversions that affect long-term acceptance of the food. Such is the importance of nutrition in cats with CKD that following up with caregivers on diet acceptance and monitoring BCS and MCS in nurse/technician clinics should be encouraged.

Diets marketed for early renal disease typically provide slightly more protein than diets marketed for more advanced disease. Ultimately, the decision on whether to feed an 'early' or 'advanced' renal diet may be based on factors other than the cat's assigned IRIS stage. For example, if a cat is eating fewer calories than ideal, then feeding a higher protein diet may be preferable to ensure the cat is meeting their protein requirement. Additionally, many of the 'early' renal diets provide slightly more phosphate and may have lower calcium:phosphate (Ca:P) ratios; however, this is not universally the case.

Further management of hyperphosphataemia

Feeding a phosphate-restricted diet is the initial treatment of choice to maintain serum phosphate concentrations within a target range. However, if feeding a renal diet is not sufficient to control serum phosphate, or the cat will not eat the veterinary therapeutic diet, phosphate binders may be mixed into all meals fed to the cat. Types of phosphate binders include aluminium hydroxide, calcium

carbonate, sevelamer and lanthanum-based products.

Management of hypercalcaemia

CKD is a cause of hypercalcaemia, and hypercalcaemia can occur when a cat is fed a phosphate-restricted diet. The consequence of ionised hypercalcaemia in cats with CKD is an increased risk of nephrocalcinosis, which appears to be associated with more progressive CKD.³² Hypercalcaemia may be managed with dietary modification and feeding a veterinary therapeutic diet with an appropriate Ca:P ratio (<1.4:1) or adding chia seeds to the diet; other measures include the administration of bisphosphonates.

Management of nausea and hyporexia

Cats with CKD can suffer from nausea, vomiting and inappetence. These clinical signs are more prevalent in cats with higher stage CKD;³³ hence, when observed in cats with IRIS stage 1 or 2 CKD, comorbidities should be considered. Inappetence is a significant quality of life concern for caregivers,³⁴ and may result in a loss of body weight and reduction in BCS and MCS in the cat.³⁵ A reduced appetite should, therefore, be actively managed as soon as such losses are appreciated, and any complications of CKD that can contribute to inappetence, such as dehydration, constipation, hypokalaemia, acidosis and anaemia, should be addressed. Centrally acting antiemetics, such as maropitant, mirtazapine and ondansetron,^{36–38} should be considered for management. The ghrelin receptor agonist capromorelin may be useful for managing appetite and body condition loss and maintaining MCS.³⁹

Management and prevention of dehydration

Dehydration is common in CKD patients due to their impaired ability to concentrate urine, and can lead to inappetence, lethargy, weakness,

constipation and an increased susceptibility to uraemic crises.³³ Cats with clinical dehydration may benefit from hospitalisation and intravenous fluid therapy to correct deficits, followed by home interventions to increase their voluntary water intake. An iCatCare resource entitled 'Increasing water intake. Encouraging your cat to drink: a guide for caregivers' (available at icatcare.org/cat-advice/cat-carer-guides) describes various ways in which water intake can be increased. These include:

- Providing free access to multiple water sources, including water fountains and wide-brimmed ceramic bowls, positioned in easily accessible locations for the cat (Figure 8a);
- Raising water bowls so that a cat with osteoarthritis does not need to lower their head to drink (Figure 8b);
- Feeding a wet diet (taking into consideration the need for a careful transition, as well as the lower calorie content per gram in wet foods);
- Adding water to the food (taking care not to adversely affect the cat's calorie or nutrient intake);



Figure 8: Interventions to increase voluntary water intake, such as (a) encouraging the use of a water fountain or (b) raising the water bowl to a more comfortable position, can benefit cats with chronic kidney disease. Courtesy of (a) Samantha Taylor and (b) Sarah Collins

- Using nutrient-enriched water products; these supplements have been shown to improve voluntary water intake and hydration in dehydrated cats.⁴⁰

Subcutaneous fluids administered at home by caregivers are not recommended for all cats with CKD, but are well accepted by most cats and, hence, can be considered for those with ongoing dehydration despite the measures discussed above having been taken (Figure 9).⁴¹ Techniques such as warming the fluids, giving treats and performing the procedure efficiently improve patient tolerance.⁴¹ Fluid is administered via a giving set and butterfly or needle.

Management of anaemia

The kidney is responsible for producing the hormone erythropoietin (EPO), which stimulates the bone marrow to produce red blood cells. As CKD progresses, EPO production decreases and anaemia can result. Consequences of anaemia include weakness, lethargy, inappetence and potential acceleration of CKD progression.^{8,42-44} Treatment may improve the cat's appetite and energy levels, and the cat's PCV may increase with the administration of injectable darbopoeitin or oral molidustat. Iron supplementation may also be required.

Management of proteinuria

Persistent proteinuria is associated with disease progression and mortality in cats with CKD.^{7,8,45,46} Patients with a persistent UPC of >0.4 (IRIS substage proteinuric) should be treated with a renal diet and pharmacological inhibition of the RAAS. Treatment options include angiotensin receptor blockers (eg, telmisartan) or angiotensin-converting enzyme inhibitors (eg, benazepril). As the treatment of hypertension can reduce proteinuria,⁴⁶ hypertensive cats should be reassessed for



Figure 9: Subcutaneous fluids can be given at home by caregivers with training and support to maintain hydration in cats with chronic kidney disease. *Courtesy of Sarah Caney*

Tip

Patient tolerance of subcutaneous fluids can be improved by warming the fluid, providing treats and ensuring that the caregiver is confident performing the procedure efficiently.



proteinuria once systolic blood pressure is <160 mmHg and prior to additional antiproteinuric medication.

Management of hypertension

Management of hypertension can prevent target organ damage and reverse or improve any existing clinical signs of target organ damage. The goal of management is to reduce systolic blood pressure to a normotensive level (ideally <140 mmHg but otherwise <160 mmHg). Options for treatment include telmisartan or amlodipine.

Management of constipation

Cats with CKD are observed to defecate less frequently than healthy cats.^{47,48} The kidneys fail to provide appropriate urine-concentrating ability and, hence, chronic subclinical

dehydration results in water being reabsorbed from the colon to compensate. Additionally, hypokalaemia, uraemic toxins, diet and the use of phosphate binders may contribute to constipation, and comorbidities such as osteoarthritis may cause a reluctance to use the litter tray. Environmental factors, including the availability/number of litter trays, their location and the litter substrate, can also influence defecation frequency. It is important to discuss with the caregiver the frequency of defecation, time spent defecating, the degree of difficulty or straining and any vomiting associated with defecation, as this information may not be volunteered but is an important part of history-taking. Also, faecal pellets left around the house may indicate dyschezia. Physical examination may reveal small hard faeces in the descending colon with a build-up of faecal material before the pelvic inlet, or, in more severe cases, a large amount of hard faecal material in the colon.

After correction of dehydration and hypokalaemia, optimising litter facilities and providing analgesia, if required, oral osmotic stool softeners can be introduced. Polyethylene glycol (PEG) 3350 (one-eighth to one-quarter of a teaspoon q12-24h) is an osmotic laxative that has been assessed in normal cats and found to be effective as a stool-softening agent.⁴⁹ Moreover, it is thought by some to be more effective than lactulose, which is processed by intestinal microflora and may lead to bloating, as well as being challenging to administer to many cats.⁴⁹ Another important consideration is that PEG 3350 comes as a powder that appears to be well tolerated. Bulk-forming agents such as psyllium (1/4 teaspoon daily) or canned pumpkin (1/2-1 teaspoon daily) may also be useful as fibre sources in the management of constipation associated with CKD in patients with

normal hydration, if accepted mixed into food.^{50,51} Proton pump inhibitors (eg, omeprazole) are a second tier of medical management for constipation associated with CKD, as the degree to which inherent colonic motility is affected has not been assessed in feline CKD patients.

Other treatments

Supplementation with potassium may be needed if deficient, and strategies to decrease the production and absorption of uraemic toxins may be prescribed.

Optimising the environment for cats with CKD

The environment in which the cat lives, both physical and social, can influence mental wellbeing and, therefore, also impact physical health.⁵² This is an area of chronic disease management that is often overlooked but where nurses and technicians can advise and support caregivers. Optimising the home environment of a cat with CKD can improve their quality of life in several ways. For example, adequate litter trays and appropriate water provision can encourage regular defecation and hydration, and an optimised environment can reduce stress and, hence, encourage food intake. Additionally, cats with CKD require diagnostic testing and monitoring, so the veterinary environment will also be impactful. Stressors associated with visits to the clinic could affect a cat's health and alter physical

Key point

The home environment should be optimised to improve the quality of life of cats with chronic kidney disease – for example, by providing adequate litter trays and multiple ways to hydrate.



examination and biochemical parameters. Additionally, caregivers observing protective (negative) emotions in their cat during a visit could discourage them from presenting them to the clinic in the future, affecting the detection of complications and limiting important interventions.^{52,53}

Home environment

Optimising the home environment for a cat with CKD can focus around

meeting the ‘five pillars’ of a healthy feline environment,⁵⁴ which include providing the following: (1) a safe space; (2) multiple and separated resources; (3) opportunities for play and performing predatory behaviour; (4) positive, consistent and predictable human-cat social interactions; and (5) an environment that respects the cat’s senses. In practical terms, a home environment can be improved for cats with CKD, as suggested in Figure 10.

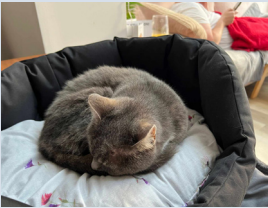
	Pillar 1: Provide a safe space Provide predictable, secure and readily accessible resting areas in multiple locations. They should be within easy reach for a cat with mobility issues or pain from degenerative joint disease (DJD), with the use of ramps or steps, as necessary, and located away from busy areas of the home. Quiet, warm, sunny resting spots with deep bedding encourage use.
	Pillar 2: Provide multiple and separated resources Cats with chronic kidney disease (CKD) may need to urinate and drink more frequently, so should have easy access to litter trays and multiple water sources. Resources should be distributed across all levels of the home and in sufficient number (at least two of each resource per cat). Food and water can be offered in raised bowls, and horizontal scratching opportunities provided for cats with DJD. Litter trays should be low sided with a soft substrate and regularly cleaned.
	Pillar 3: Provide opportunity to play and perform predatory behaviour Even older cats with chronic illness will benefit from activity. Analgesia may help cats with DJD engage with light activity and low-impact play, which can also help trigger engaging emotions and prevent constipation. Cats could be encouraged to walk around a garden or engage with a fishing toy for short periods. Incorporating toys with auditory or olfactory stimuli (eg, crinkle material, catnip, valerian, silvervine), playing at predictable times and locations, using short, consistent sessions, and offering cardboard boxes placed on their sides supports safe, comfortable engagement.
	Pillar 4: Provide positive, consistent and predictable human-cat social interaction Allow cats to choose the type and duration of interaction and never force handling. Some cats with chronic illness may groom less and enjoy being brushed; others may still have the appetite for occasional treats from their caregivers. All family members should be educated not to pick up the cat, but approach slowly with a soft voice to prevent startling. The cat should be allowed to disengage from interactions with people at their choosing.
	Pillar 5: Provide an environment that respects the importance of the cat’s senses Strong scents should be avoided (perfumes, air fresheners) and use of feline pheromone products considered. Food could be warmed to increase aromas and encourage eating. Sudden loud noises should be avoided to prevent startling the cat.

Figure 10: Optimising the home environment for a cat with chronic kidney disease using the ‘five pillars’ model as a guide. Modified from the 2026 iCatCare consensus guidelines on the diagnosis and management of chronic kidney disease in cats. *J Feline Med Surg*. Available at: <https://journals.sagepub.com/doi/full/10.1177/1098612X261466553>. Copyright The Authors 2026. Images courtesy of Samantha Taylor (Pillars 1, 2 and 5), International Cat Care (Pillar 3) and Sarah Caney (Pillar 4)

Veterinary environment

The clinic environment can be distressing for any cat, and older cats, such as those with CKD, may be more vulnerable to anxiety caused by leaving the home environment. They may have concurrent disease (eg, osteoarthritis, hypertension) resulting in pain, which can increase protective emotions.⁵⁵ A comprehensive discussion on improving the veterinary visit experience for cats is provided in the ISFM/AAFP cat friendly guidelines, which cover both veterinary interactions⁵⁶ and the veterinary environment,⁵⁷ and are available at bit.ly/JFMSCatFriendly. Key principles from these pivotal resources include:

- **Species separation** – ideally, having feline-only waiting areas, wards and consulting rooms. Where this is not feasible, allowing cats to wait in the car and/or offering senior/feline-only clinic consulting periods.
- **Cat friendly interactions** – gentle handling and avoidance of restraint that could cause pain (eg, elbow extension for cephalic blood sampling). Examination should be conducted in comfortable positions and in locations of the patient's choosing (eg, in the base of the cat carrier or on a soft bed).
- **Provision of analgesia** – prior to examinations and procedures.
- **Pre-visit anxiolysis** – use of gabapentin or pregabalin pre-visit to reduce anxiety. Note that a lower dosage of these drugs may be preferred in cats with CKD.
- **Blood sampling considerations** – use of local anaesthetic creams for blood sampling, use of treats as distraction (Figure 11) and taking only the minimum volume of blood required for necessary tests, to avoid iatrogenic lowering of PCV.
- **Preparation of equipment** – having all equipment (eg, blood pressure machine, blood sampling kit, including quiet clippers) ready in



Figure 11: Treats can be used as a distraction for some cats to allow blood sampling. *Courtesy of Jessica Quimby*



Figure 12: All hospitalised cats can benefit from a place to hide, even a cardboard box with a comfortable base. *Courtesy of Sam Taylor*

the consulting room to avoid going in and out.

- **Ward environment** – making the ward cat friendly with a warm ambient temperature and the provision of accessible resources close to the cat, including low-sided litter trays and somewhere to hide (Figure 12); also minimising hospitalisation periods, where possible.
- **Dietary considerations** – taking a nutritional history and providing the cat with their familiar food while in hospital. Note that it is important not to introduce a new diet, such as a renal diet, to a cat who is unwell and hospitalised.

The practice team – including veterinarians, reception staff, nurses/technicians and patient care assistants – should all be familiar

with cat friendly principles. Clinic meetings to discuss how to optimise the clinic environment for senior cats can be helpful.

Addressing the caregiver burden of CKD

It is important to recognise the common and often substantial caregiver burden associated with caring for a cat with CKD.⁵⁸ Caregivers often may feel anxious about the prognosis for, and quality of life of, their cat, as well as their ability to provide medications.

Client communication around feline CKD should focus on the individual cat and the caregiver's unique needs, involving caregivers as much as they wish and adopting a non-judgemental, empathetic and shared decision-making approach.

Clinicians should:

- Consider this emotional strain in consultations and validate caregivers' worries;
- Provide clear, ongoing communication about CKD progression, expectations and management goals. Caregivers appreciate being well informed about their cat's illness;¹⁶
- Involve nurses/technicians in communicating with caregivers to provide practical information on administering medications, introducing dietary changes and monitoring cats with CKD;
- Encourage involvement of caregivers in their cat's care – for example, by keeping diaries and monitoring body weight at home (Figure 13).

Actively addressing the burden of caring for a cat with CKD improves caregiver compliance, supports long-term CKD management, and enhances outcomes for both cats and their families.

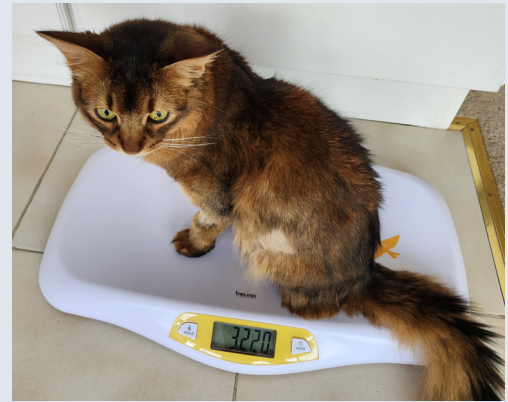


Figure 13: Weighing cats at home and communicating results to the veterinary clinic can assist monitoring the cat with chronic kidney disease. *Courtesy of Sam Taylor*

Medicating cats with CKD: strategies for improving caregiver compliance

Ultimately, caring for a patient with CKD is a balance between providing necessary therapies and preserving quality of life. A management plan should be tailored to the individual patient based on the caregiver's abilities and the cat's level of acceptance. Education regarding the rationale for, and evidence behind, prescribed therapies may improve compliance.¹⁶

Tips for medicating cats with CKD include:

- Prioritising the most important medications for cats who are challenging to medicate;
- Researching and discussing options for different medication formulations (liquids, tablets, powders, capsules) based on individual cat and caregiver preferences;
- Informing caregivers about the practicalities of giving the medication; for example, administering in food (liquid treats, pill 'putty' products), the option to crush tablets or open capsules;
- Demonstrating how to give the medication directly into the mouth, if necessary, without causing the cat discomfort;
- For painful cats, considering administering injectable analgesics initially at the veterinary clinic to improve comfort and help facilitate ongoing oral medication administration;
- Avoiding bitter-tasting medications or those requiring a large volume of liquid or large tablet size;
- Avoiding hiding medication in the cat's main meals, as this can cause food aversion; the exception to this is phosphate binders, which should ideally be mixed with the cat's food, but could be given after eating;
- Combining medications in a gel capsule, so only a single administration is required.

Following up with caregivers via their preferred communication route (telephone, text, email) allows challenges to be discussed and alternatives to be found, if necessary.

Anaesthesia in cats with CKD

Nurses and technicians are likely to be involved in the anaesthesia management of cats with CKD. While a comprehensive discussion of anaesthesia in the cat with CKD is beyond the scope of this article, fundamental considerations include the cat's current clinical status as well as the influence of reduced renal function on drug pharmacokinetics. Anaesthesia and analgesia should not be withheld in cats with CKD when indicated, but it does require special consideration. The goal is to preserve renal perfusion and oxygen delivery, avoid hypotension and dehydration, and minimise changes in GFR that could lead to additional renal damage. Priorities include:

- IRIS staging and substaging prior to anaesthesia so that the stage and severity of renal dysfunction is understood. Preoperative blood pressure should be recorded;
- Measurement of PCV and electrolytes shortly before the procedure;
- Correction of dehydration and electrolyte abnormalities prior to anaesthesia. Caregivers may be asked to bring their cat in earlier for intravenous fluid therapy to restore circulating volume and support haemodynamics prior to the procedure;
- Maintaining the cat's access to water up to premedication and limiting fasting to 4–6 h (after, ideally, feeding one-quarter of the daily food requirement as wet food);
- Considering the potential for prolonged activity of drugs undergoing renal elimination and adjusting dosage intervals accordingly. For example, ketamine is excreted unchanged in cats and may accumulate in patients with CKD;
- Keeping in mind that some premedication agents, such as

Key point

Anaesthesia in cats with chronic kidney disease should not be withheld; however, special consideration is required, with the goal being to preserve renal perfusion and oxygen delivery, avoid hypotension and dehydration, and minimise changes in the glomerular filtration rate.



acepromazine or α_2 -agonists, may cause vasodilation, hypotension and/or a reduction in cardiac output or GFR, particularly when combined with volatile anaesthetics;

- Titrating drugs for induction and maintenance of anaesthesia to avoid cardiovascular depression, and optimising locoregional analgesic techniques to reduce the dosages needed;
- Monitoring mean arterial pressure and maintaining it at or above 70 mmHg to ensure adequate renal perfusion. Downward trends in blood pressure should be promptly managed by decreasing inhalational anaesthesia, instituting bolus fluid therapy or using vasopressor/positive inotropic agents, when needed. Planning interventions in advance of anaesthesia can facilitate prompt responses;
- Providing fluid therapy with crystalloids (eg, Ringer's solution) on an individual patient basis to restore and maintain an effective circulating volume while avoiding dehydration and fluid overload; an appropriate fluid therapy rate is usually around 3–4 ml/kg/h. Reassessment of hydration status and renal parameters may be considered after surgery,

particularly with higher risk procedures. Cats with CKD are at risk of fluid deficits but, importantly, also fluid overload due to impaired fluid excretion, so monitoring and adjustment is recommended;

- Maintaining normothermia intraoperatively and during recovery;
- Avoiding non-steroidal anti-inflammatory drugs (NSAIDs), where possible. Alternative analgesics and use of locoregional blocks, for example, should be utilised and pain-scoring systems implemented. If NSAIDs are considered, CKD must be stable, bleeding should not have occurred during surgery, and the cat should be conscious, eating, drinking or receiving fluids so that return of renal blood flow autoregulation is confirmed; and
- Encouraging a rapid return to voluntary food and water intake postanaesthesia (providing analgesia, antiemetics and appetite stimulants, if required).

Seeking advice from a veterinary anaesthesia specialist may be beneficial in cases where long or complex procedures are required in cats with CKD.

Monitoring the cat with CKD: a nurse's role

Monitoring checks for cats with CKD are aimed at understanding patient progress (ie, improvements and deteriorations), compliance challenges and other caregiver support needs, and should be individualised in the light of cat and caregiver factors such as temperament and financial limitations. Nurses and technicians can play a key role in maintaining this contact, monitoring the cat's progress, and communicating with, and providing support to, caregivers of cats with CKD; for example, by:

- Obtaining recent history from caregivers; for example, on appetite, demeanour and litter tray use;
- Holding nursing consultations to weigh the cat and monitor their BCS and MCS;
- Measuring blood pressure;
- Supporting in diet transition and then assessing compliance to, and suitability of, the current diet;
- Following up to support with medication compliance;
- Assessing caregiver records; for example, via questionnaires;
- Triaging cats with CKD if any problems are reported or a deterioration in clinical condition has been detected;
- Training caregivers in supportive procedures such as the administration of subcutaneous fluid therapy in higher stage CKD patients; and
- Discussing palliative care or end-of-life considerations and performing quality of life assessment.

Following the initial diagnosis of CKD, or any significant changes in the cat's clinical condition, it can be helpful to assess/reassess the cat's progress relatively frequently; for example, every 2–4 weeks, depending on the patient's IRIS stage and the severity of their clinical signs. Thereafter, frequency of check-ups will depend on the cat's progress; for example, cats with stable CKD may only require reassessment every 3–6 months.

Caregiver education and training often facilitate caregivers becoming valuable contributors to clinical insight; for example, by providing information on their cat's appetite, toileting behaviour and hydration status. Encouraging the use of clumping cat litter can allow more precise monitoring of urination patterns, while monitoring defecation frequency and faecal consistency can help identify poor hydration and

hypokalaemia, as dehydrated or hypokalaemic cats tend to defecate less often and pass hard, dry faeces.

Prognosis for cats with CKD

The prognosis for cats with CKD is highly variable and unpredictable. Many cats diagnosed with CKD and managed with appropriate supportive care remain stable and well for years. Tracking progress in as much detail as possible assists in identifying cats whose disease is progressing at a faster rate and, in turn, advising and supporting caregivers in this situation. Stability vs progression influences survival time, with cats with stable CKD having a median survival time of 894 days (95% confidence interval: 812–1052) and cats with progressive CKD having a median survival time of 287 days (95% confidence interval: 246–354).¹¹

Conclusions

Nurses and technicians should be involved in the team approach to care of the cat with CKD. Detecting cats with CKD via senior health assessments and carrying out sampling, monitoring, caregiver communication and support, and care in the clinic when cats are hospitalised are just some of the areas in which nurses and technicians can make an important difference in managing cats with this common condition.

References

- 1 Sparkes AH, Caney S, Chalhoub S, et al. **ISFM consensus guidelines on the diagnosis and management of feline chronic kidney disease.** *J Feline Med Surg* 2016; 18: 219–239.
- 2 Brown CA, Elliott J, Schmiedt CW, et al. **Chronic kidney disease in aged cats: clinical features, morphology, and proposed pathogenesis.** *Vet Pathol* 2016; 53: 309–326.
- 3 Marino CL, Lascelles BD, Vaden SL, et al. **Prevalence and classification of chronic kidney disease in cats randomly selected from four age groups and in cats recruited for degenerative joint disease studies.** *J Feline Med Surg* 2014; 16: 465–472.
- 4 Miele A, Sordo L and Gunn-Moore DA. **Feline aging: promoting physiologic and emotional well-being.** *Vet Clin North Am Small Anim Pract* 2020; 50: 719–748.
- 5 Bartges JW. **Chronic kidney disease in dogs and cats.** *Vet Clin North Am Small Anim Pract* 2012; 42: 669–692.
- 6 DiBartola SP, Rutgers HC, Zack PM, et al. **Clinicopathologic findings associated with chronic renal disease in cats: 74 cases (1973–1984).** *J Am Vet Med Assoc* 1987; 190: 1196–1202.
- 7 Syme HM, Markwell PJ, Pfeiffer D, et al. **Survival of cats with naturally occurring chronic renal failure is related to severity of proteinuria.** *J Vet Intern Med* 2006; 20: 528–535.
- 8 Chakrabarti S, Syme HM and Elliott J. **Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease.** *J Vet Intern Med* 2012; 26: 275–281.
- 9 Boyd LM, Langston C, Thompson K, et al. **Survival in cats with naturally occurring chronic kidney disease (2000–2002).** *J Vet Intern Med* 2008; 22: 1111–1117.
- 10 Geddes RF, Elliott J and Syme HM. **Relationship between plasma fibroblast growth factor-23 concentration and survival time in cats with chronic kidney disease.** *J Vet Intern Med* 2015; 29: 1494–1501.
- 11 Tang PK, Jepson RE, Chang YM, et al. **Elevation of the slope of inverse creatinine to define progressive chronic kidney disease in cats and associated risk factors.** *Vet J* 2026; 317. DOI: 10.1016/j.tvjl.2026.106659.
- 12 Chen CN, Chou CC, Tsai PSJ, et al. **Plasma indoxyl sulfate concentration predicts progression of chronic kidney disease in dogs and cats.** *Vet J* 2018; 232: 33–39.
- 13 Nakhoul GN, Huang H, Arrigain S, et al. **Serum potassium, end-stage renal disease and mortality in chronic kidney disease.** *Am J Nephrol* 2015; 41: 456–463.
- 14 Caney SM. **An online survey of dietary and phosphate binder practices of owners of cats with chronic kidney disease.** *J Feline Med Surg* 2017; 19: 1040–1047.
- 15 Elliott J and Barber PJ. **Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995.** *J Small Anim Pract* 1998; 39: 78–85.
- 16 Magalhaes TR, Lourenco AL, Corbee RJ, et al. **Caregivers' perspectives on feline chronic kidney disease in Portugal: a questionnaire-based study.** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251377486.
- 17 Dowgray N, Pinchbeck G, Eyre K, et al. **Aging in cats: owner observations and clinical finding in 206 mature cats at enrolment to the cat prospective aging and welfare study.** *Front Vet Sci* 2022; 9. DOI: 10.3389/fvets.2022.859041.
- 18 Mortier F, Daminet S, Marynissen S, et al. **Value of repeated health screening in 259 apparently healthy mature adult and senior cats followed for 2 years.** *J Vet Intern Med* 2024; 38: 2089–2098.
- 19 Freeman LM, Lachaud MP, Matthews S, et al. **Evaluation of weight loss over time in cats with chronic kidney disease.** *J Vet Intern Med* 2016; 30: 1661–1666.
- 20 Brusach K, Lorbach S, Quimby J, et al. **Measurement of ghrelin as a marker of appetite dysregulation in cats with and**

- without chronic kidney disease. *Vet Sci* 2023; 10. DOI: 10.3390/vetsci10070464.
- 21 Karpozilou A, De Stefani A and Liatis T. **Cervical ventroflexion in cats: 86 cases (2003–2024).** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251348328.
- 22 Dorsch R, Teichmann-Knorrn S and Sjetne Lund H. **Urinary tract infection and subclinical bacteriuria in cats: a clinical update.** *J Feline Med Surg* 2019; 21: 1023–1038.
- 23 Benjamin SE and Drobatz KJ. **Retrospective evaluation of risk factors and treatment outcome predictors in cats presenting to the emergency room for constipation.** *J Feline Med Surg* 2020; 22: 153–160.
- 24 Finch N. **Measurement of glomerular filtration rate in cats: methods and advantages over routine markers of renal function.** *J Feline Med Surg* 2014; 16: 736–748.
- 25 Michael H, Szlosek D, Clements C, et al. **Symmetrical dimethylarginine: evaluating chronic kidney disease in the era of multiple kidney biomarkers.** *Vet Clin North Am Small Anim Pract* 2022; 52: 609–629.
- 26 Finco DR, Adams DD, Crowell WA, et al. **Food and water intake and urine composition in cats: influence of continuous versus periodic feeding.** *Am J Vet Res* 1986; 47: 1638–1642.
- 27 Rishniw M and Bicalho R. **Factors affecting urine specific gravity in apparently healthy cats presenting to first opinion practice for routine evaluation.** *J Feline Med Surg* 2015; 17: 329–337.
- 28 Cordella A, Lenz J, Bertal M, et al. **Ultrasonographic features of feline renal neoplasia: a retrospective study on 187 cases.** *J Feline Med Surg* 2026; 28. DOI: 10.1177/1098612X261432299.
- 29 Griffin S. **Feline abdominal ultrasonography: what's normal? what's abnormal? The kidneys and perinephric space.** *J Feline Med Surg* 2020; 22: 409–427.
- 30 Elliott J, Rawlings JM, Markwell PJ, et al. **Survival of cats with naturally occurring chronic renal failure: effect of dietary management.** *J Small Anim Pract* 2000; 41: 235–242.
- 31 Ross SJ, Osborne CA, Kirk CA, et al. **Clinical evaluation of dietary modification for treatment of spontaneous chronic kidney disease in cats.** *J Am Vet Med Assoc* 2006; 229: 949–957.
- 32 Tang PK, Geddes RF, Chang YM, et al. **Risk factors and implications associated with ultrasound-diagnosed nephrocalcinosis in cats with chronic kidney disease.** *J Vet Intern Med* 2024; 38: 1563–1576.
- 33 Quimby J. **Chronic kidney disease.** In: Ettinger SJ, Feldman EC and Côté E (eds). *Ettinger's textbook of veterinary internal medicine*. 9th ed. Elsevier, 2024, pp 2089–2106.
- 34 Lorbach SK, Quimby JM, Nijveldt E, et al. **Evaluation of health-related quality of life in cats with chronic kidney disease.** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251367535.
- 35 Taylor S, Chan DL, Villaverde C, et al. **2022 ISFM consensus guidelines on management of the inappetent hospitalised cat.** *J Feline Med Surg* 2022; 24: 614–640.
- 36 Quimby JM, Lake RC, Hansen RJ, et al. **Oral, subcutaneous, and intravenous pharmacokinetics of ondansetron in healthy cats.** *J Vet Pharmacol Ther* 2014; 37: 348–353.
- 37 Quimby JM and Lunn KF. **Mirtazapine as an appetite stimulant and anti-emetic in cats with chronic kidney disease: a masked placebo-controlled crossover clinical trial.** *Vet J* 2013; 197: 651–655.
- 38 Quimby JM, Brock WT, Moses K, et al. **Chronic use of maropitant for the management of vomiting and inappetence in cats with chronic kidney disease: a blinded, placebo-controlled clinical trial.** *J Feline Med Surg* 2015; 17: 692–697.
- 39 Wofford JA, Milliken MacKinnon A and Heinen E. **Capromorelin promotes weight gain in cats with unintended weight loss: a randomized, masked, placebo-controlled clinical trial.** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251379924.
- 40 Peralta M, Nichelason A and Trepanier L. **Voluntary acceptance of nutrient-enriched water supplement and promotion of water intake in clinically dehydrated cats.** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251334279.
- 41 Cooley CM, Quimby JM, Caney SM, et al. **Survey of owner subcutaneous fluid practices in cats with chronic kidney disease.** *J Feline Med Surg* 2018; 20: 884–890.
- 42 Elliott J. **Therapeutics of managing reduced red cell mass associated with chronic kidney disease – is there a case for earlier intervention?** *J Vet Pharmacol Ther* 2023; 46: 145–157.
- 43 Spencer S, Wheeler-Jones C and Elliott J. **Hypoxia and chronic kidney disease: possible mechanisms, therapeutic targets, and relevance to cats.** *Vet J* 2021; 274. DOI: 10.1016/j.tvjl.2021.105714.
- 44 Nangaku M and Eckardt KU. **Pathogenesis of renal anemia.** *Semin Nephrol* 2006; 26: 261–268.
- 45 King JN, Tasker S, Gunn-Moore DA, et al. **Prognostic factors in cats with chronic kidney disease.** *J Vet Intern Med* 2007; 21: 906–916.
- 46 Jepson RE, Elliott J, Brodbelt D, et al. **Effect of control of systolic blood pressure on survival in cats with systemic hypertension.** *J Vet Intern Med* 2007; 21: 402–409.
- 47 Jones SE, Quimby JM, Summers SC, et al. **Survey of defecation habits in apparently healthy and chronic kidney disease cats.** *J Feline Med Surg* 2022; 24: 131–141.
- 48 George ZM, Quimby JM, Jones S, et al. **Quantification of defecation frequency in cats with and without chronic kidney disease.** *J Feline Med Surg* 2025; 27. DOI: 10.1177/1098612X251348011.
- 49 Tam FM, Carr AP and Myers SL. **Safety and palatability of polyethylene glycol 3350 as an oral laxative in cats.** *J Feline Med Surg* 2011; 13: 694–697.
- 50 Freiche V, Houston D, Weese H, et al. **Uncontrolled study assessing the impact of a psyllium-enriched extruded dry diet on faecal consistency in cats with constipation.** *J Feline Med Surg* 2011; 13: 903–911.
- 51 Rudinsky AJ and Parker VJ. **Therapeutic applications of dietary fiber: managing fiber-responsive gastrointestinal conditions in dogs and cats.** *Vet Clin North Am Small Anim Pract* 2026; 56: 603–617.
- 52 Heath S. **Environment and feline health: at home and in the clinic.** *Vet Clin North Am Small Anim Pract* 2020; 50: 663–693.
- 53 Stella J, Cronley C and Buffington T. **Effects of stressors on the behavior and physiology of domestic cats.** *Appl Anim Behav Sci* 2013; 143: 157–163.

- 54 Ellis SL, Rodan I, Carney HC, et al. **AAFP and ISFM feline environmental needs guidelines.** *J Feline Med Surg* 2013; 15: 219–230.
- 55 Heath S, Dowgray N, Rodan I, et al. **10 years of Cat Friendly: a new model and terminology for understanding feline emotions.** *J Feline Med Surg* 2022; 24: 934–935.
- 56 Rodan I, Dowgray N, Carney HC, et al. **2022 AAFP/ISFM cat friendly veterinary interaction guidelines: approach and handling techniques.** *J Feline Med Surg* 2022; 24: 1093–1132.
- 57 Taylor S, St Denis K, Collins S, et al. **2022 ISFM/AAFP cat friendly veterinary environment guidelines.** *J Feline Med Surg* 2022; 24: 1133–1163.
- 58 Lim L, Quimby J and Caney S. **Feline chronic kidney disease is associated with significant caregiver burden.** 2024 ACVIM forum research abstract program. *J Vet Intern Med* 2024; 38: 2954–2955.